# Project 2

CS 4033/5033 Machine Learning Fundamentals, Fall 2026

**Due: Monday, November 2nd, 11:59 pm**

## 1. Motivation

Regression is one of the most important types of machine learning (ML) problems. Linear regression is one of the most prominent, best understood, and most widely used statistical approaches to regression. Therefore, knowing how to conduct linear regression is a fundamental skill for ML practitioners. In this programming assignment, you will conduct simple and multiple linear regression with one to three predictors (independent variables) and one dependent variable.

## 2. Goal

The goals of this assignment are:

- to give you baseline experience with regression,
- to give you experience in particular with simple and multiple linear regression,
- to familiarize you with some of the metrics appropriate for evaluating regression models, and
- to give you additional experience with supervised learning problems.

## 3. Assignment Overview and Data

You will conduct linear regression on nine simple synthetic datasets, each containing 100 data items. These are found in the provided comma-separated value (csv) files.

There are three levels of complexity to the datasets: Simple Linear Regression, Multiple Linear Regression with Two Independent Variables, and Multiple Linear Regression with Three Independent Variables. Within each level of complexity, there are three files of data, each generated from a different linear equation with different levels of Gaussian noise in the output dimension, as follows:

- Low noise (denoted "L" in the file name).
- Moderate noise (denoted "M" in the file name).
- High noise (denoted "H" in the file name).

### Simple Linear Regression

The first set of three data files is for simple linear regression, which means that there is one input dimension (that is, one independent variable) and one output dimension (that is, one dependent variable). The two columns in each file give the input and output dimension (that is, the independent variable $x$ and the dependent variable $y$) in that order, with a different data item in each row. The three data files are:

1. 1D L.csv
2. 1D M.csv
3. 1D H.csv

### Multiple Linear Regression with Two Independent Variables

The second set of three data files is for multiple linear regression with two independent variables, which means that there are two input dimensions. We still only have one output dimension (that is, one dependent variable) because we are doing multiple linear regression, rather than multivariate regression. The three columns in each file give the two input dimensions and one output dimension (that is, the independent variables $x_1$ and $x_2$, and the dependent variable $y$) in that order, with a different data item in each row. The three data files are:

1. 2D L.csv
2. 2D M.csv
3. 2D H.csv

### Multiple Linear Regression with Three Independent Variables

The third set of three data files is for multiple linear regression with three independent variables, which means that there are three input dimensions. Again, we still only have one output dimension (that is, one dependent variable) because we are doing multiple linear regression, rather than multivariate regression. The four columns in each file give the three input dimensions and one output dimension (that is, the independent variables $x_1$, $x_2$, and $x_3$, and the dependent variable $y$) in that order, with a different data item in each row. The three data files are:

1. 3D L.csv
2. 3D M.csv
3. 3D H.csv

## 4. Assignment Steps

Carry out the following steps. Underlined steps require a written response, those in **bold** require you to **write** or obtain and justify software, and those in *italics* require you to collect data. Written responses, **code**, and *data* will be turned in for grading. Note that the steps below refer to a "regression model" in the singular, although you will fit a separate linear regression model to each dataset.

1. List the form of the equation you are attempting to model for each dataset.
2. List and justify the metrics you will use to measure and report on the performance quality of regression you accomplish with each dataset.
3. List and justify any graphical representations you will use to understand the data itself and the regression performance.
4. **Implement regression code for your project.** Alternatively, you may use existing linear regression software or services. If you choose this alternative, describe and justify your choice of the software/services. Note that only software/services readily available to all OU students (including your TA) may be selected.
5. *For each dataset, conduct linear regression and collect performance data.*

6. Report your results, analysis, and conclusions regarding the application of regression to each dataset individually and in combination. Be sure to consider how performance varies as the number of input dimensions (independent variables) increases, as well as with increasing levels of noise. Include each equation determined via regression and any graphics you selected to support your analysis.

## 5. What to Turn In

### 1. Write Up

You will turn in an electronic copy of your write-up. Your write-up should be a coherent document that covers all of the underlined steps from the assignment above. Note that selected data in a digested form (such as tables or graphs) should be included in your write-up; however, your raw data should not be included here.

### 2. Code

You will turn in an electronic copy of your code. You will turn in the source code you have written for regression. Your source code should be well-structured and well-commented. It should conform to good coding standards (e.g., no memory leaks). If you selected to use existing software, provide a link to it.

### 3. Data

You will turn in an electronic copy of your data. This may be in a single file or multiple files. You will also need to include a brief write-up on how the data is organized.

## 6. Notes on this Assignment

Be sure to use proper terminology.

You may write your program from scratch or may start from programs for which the source code is freely available (such as on the web, from friends, or from student organizations). If you do not start from scratch, you **must** give a complete and accurate accounting of where all of your code came from and indicate which parts are original or changed, and which you got from which other source.

As an alternative to writing your own code, either from scratch or building on existing code, you may complete this assignment by using one of the many regression packages available these days. However, it has to be something freely available and needs to allow the configuration you've used to be exported, so that our TA and I can download it and try out the configuration you've submitted. In addition, if you use a third-party package, you **must** graph the fitted model against the data: the regression line for the one-variable datasets, the regression plane for the two-variable datasets, and, for the three-variable datasets, plots of predicted versus actual values and of the residuals.

For the written components of this assignment, you may follow the format or content of other written works, but you **must** give a complete and accurate accounting of who deserves credit for all parts of your documents.

Failure to give credit where credit is due is academic fraud and will be dealt with accordingly. See the University's web pages on academic integrity.